Tourism Pressure Across Portugal

WORKFLOW TOOLS

WHY STUDY TOURISM?

Tourism is a key driver of the Portuguese economy, contributing approximately **€34 billion in 2024** (INE, 2025), representing about **12% of GDP**. Despite its importance, its growth is not evenly distributed across regions.

WHAT IS THE PROBLEM?

Tourism demand tends to concentrate in specific regions, creating situations of high tourist pressure. In some cases, this demand is not matched by the available accommodation capacity, leading to potential **imbalances between supply and demand**.

WHAT DO WE WANT TO UNDERSTAND?

This study aims to analyse how tourism has evolved over time, how it is distributed across regions, and whether the available accommodation supply is aligned with tourist demand, identifying areas with higher pressure.

WHY DOES THIS ANALYSIS MATTER?

Understanding these patterns is essential to support **better planning decisions**, promote a more balanced distribution of tourism, and contribute to a more sustainable development of the sector.

WHO BENEFITS FROM THIS?

- Local authorities and public entities
- Real estate sector
- Accommodation and hospitality businesses
- Urban and environmental planning organizations

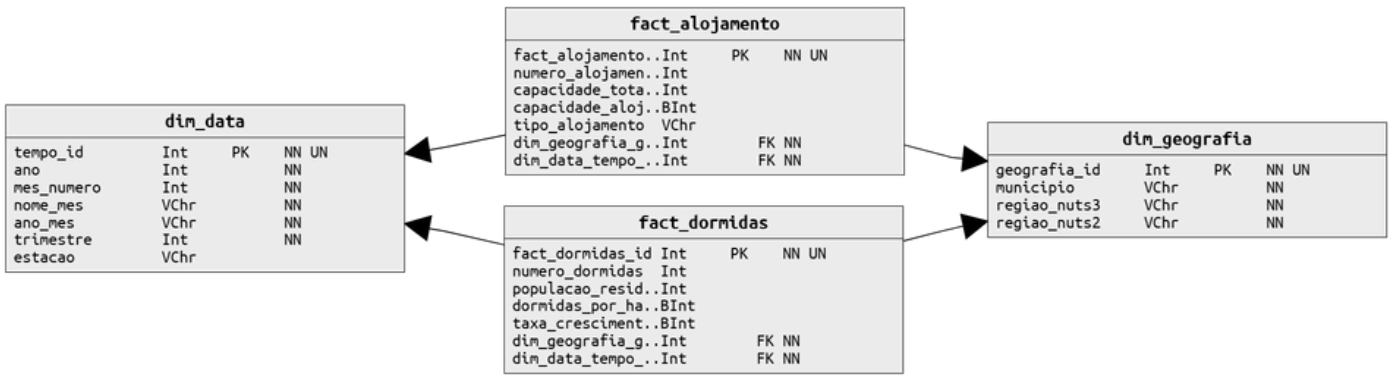
DATA SOURCES

- **INE (Instituto Nacional de Estatística) -** demographic and tourism indicators
- **TravelBI (Turismo de Portugal)** – detailed tourism activity data

The dataset covers the period **2014–2024**, with monthly granularity and geographical detail at NUTS II, NUTS III and municipality levels.

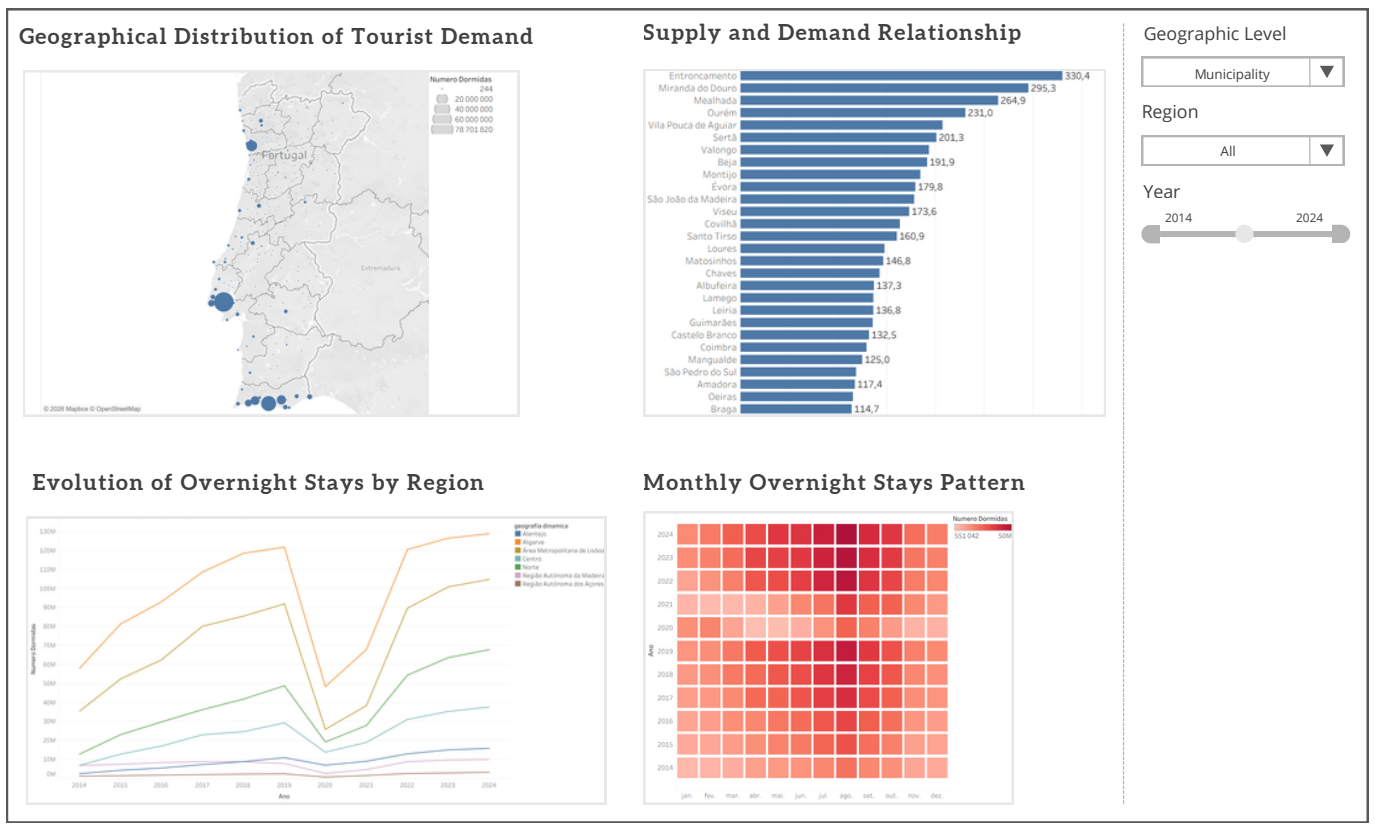
DATA WAREHOUSE FRAMEWORK

Data from **INE** and **TravelBI** were extracted and integrated, ensuring consistency across geographic levels and **monthly time granularity**. During the transformation process, **variables were standardized, temporal fields were created**, and datasets were aligned using common geographic and time keys. Additional indicators, such as **tourism pressure** and **supply-demand ratios**, were created during the transformation stage.



A **star schema** was implemented with **two fact tables**: one for **tourism demand (overnight stays)** and another for **accommodation supply (capacity and number of units)**, both linked through shared **time and geography dimensions**.

PLATAFORM



The user interface is implemented as an interactive dashboard in Tableau, combining visualisations that support the analysis of temporal evolution, geographical distribution, seasonal patterns and the relationship between tourism demand and accommodation supply. These visualisations address key questions such as how overnight stays evolve, how tourism is distributed across regions, and whether supply is aligned with demand.

Interaction with the data is enabled through filters for year, region and geographic level (NUTS II, NUTS III and Municipality), allowing analysis at different levels of detail. The dashboard supports time-based and location-based exploration through drill-down and roll-up operations, enabling both **descriptive** and **diagnostic** analysis of trends and regional imbalances.

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